



DESIGN REVIEW MEETING #1

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Zhang

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FABRICATION





BACKGROUND INFORMATION

What is the problem?

Children often leave behind craft materials and litter in classrooms, community centers, and public spaces.

Why is this a problem?

- Waste creates aesthetic concerns
- Manual cleanup can be time-consuming and inefficient.
- Craft waste is bulky and fills trash and recycling bins quickly



BACKGROUND INFORMATION

How to solve?

A system that can automatically collect and compact many small amounts of litter.

Who is this for?

- Elementary schools, community centers, daycare facilities, and possibly homes
- Teachers, custodial staff, and parents

Why do build this? Where did it come from?

Inspired by WALL-E, an autonomous robot that collects, compresses, and transports waste to clean up earth while humans relax.



NEED STATEMENT

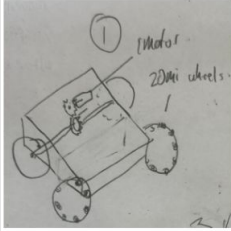
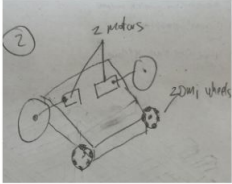
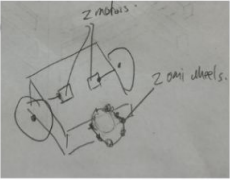
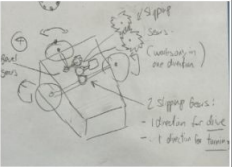
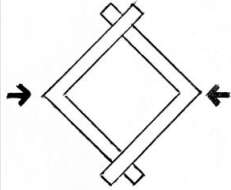
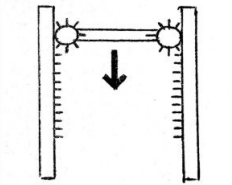
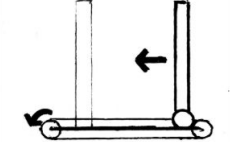
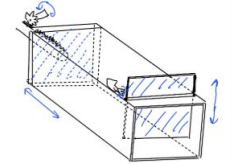
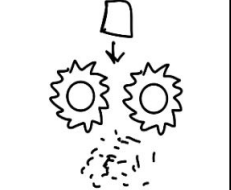
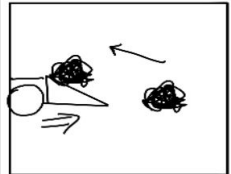
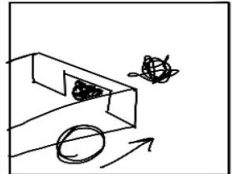
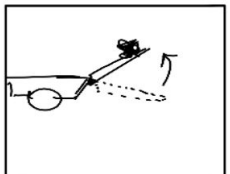
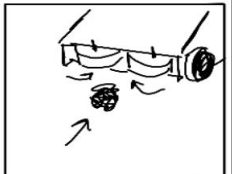
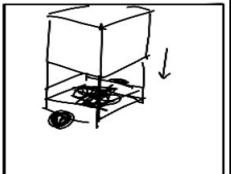
A need exists for trash to be collected, compressed, and transported to a central collection area while using only electrical components from a single VEX kit (and an additional motor).



DESIGN SPECIFICATIONS

No.	Characteristic	Relation	Value	Units	Verification Method	Comments
1	Mass (Collected Trash)	>=	10	grams	Test	Trash will be scattered where the collected amount will be weighed on a scale.
2	Volume reduction (After compression)	>=	35	%	Test	By measuring the width, length, and height of the cube before and after the compression, we can compare the volume of litter before and after and calculate % of volume reduced.
3	Vote (Aesthetics)	>=	70	%	Test	Survey a minimum of 10 peers who have watched Wall-E and ask them if they can see the resemblance.
4	Price	<=	50	CAD	Analysis	Calculate the total cost for the 3D printed parts (based on volume of model and cost per volume of filament), crafted parts, and bought parts.

CONCEPTUAL DESIGNS

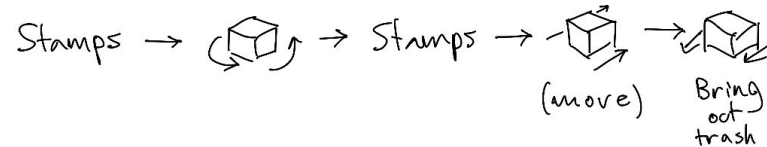
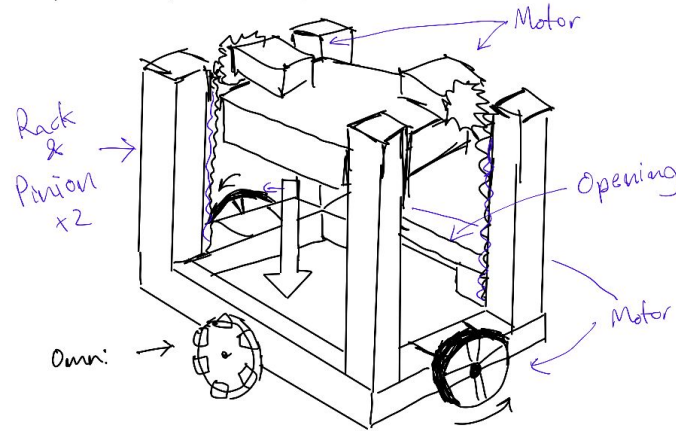
	1	2	3	4	5
Movement (How Robot gets to Trash)					67
Compression (Crush Trash)					
Lifting / Placing Trash					

CONCEPTUAL DESIGNS (DATUM)

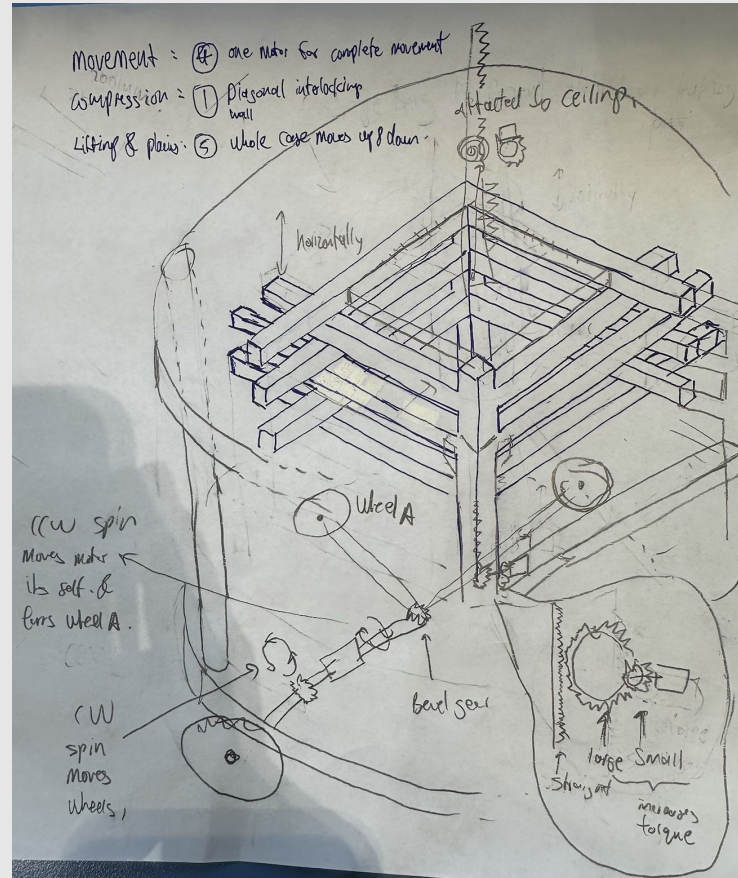
Movement (3) - 2 motor + omni

Compression (2) - Rack & pinion

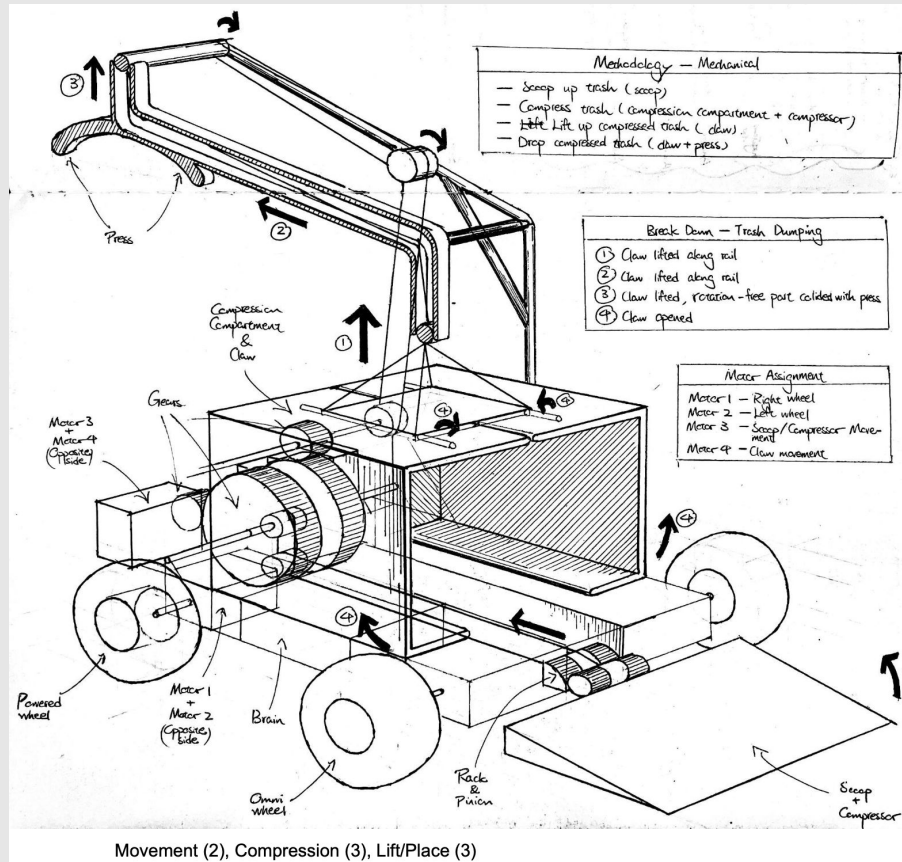
Lift/Place (2) - Opening



CONCEPTUAL DESIGNS #2



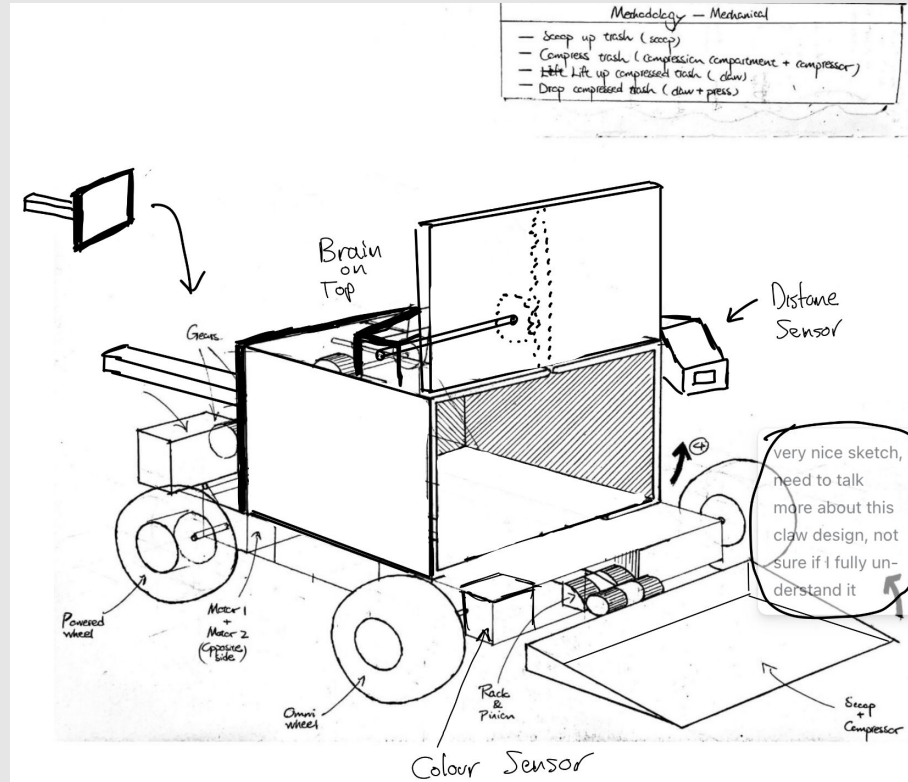
CONCEPTUAL DESIGNS #3



DECISION-MAKING MATRIX

Criteria	Datum (#1)	Concept 2	Concept 3
Safety	0	0	0
Speed	0	-1	1
Volume Reduction	0	1	0
Adaptability	0	0	1
Cost (Parts)	0	-1	-1
Efficiency	0	-1	-1
Volume	0	1	1
Total	0	-1	1

CONCEPTUAL DESIGNS #3





Fabrication

- General frame structure will be made out of VEX parts
- Dimensions of the VEX parts will be taken as the basis for additional custom parts
- Custom parts will either be 3D printed or laser cutted

The Scooper & Distance Sensor Holder

3D printing Traits:

- Customized components
- Irregular geometries

Where? Own access to printers

Compression Chamber/Cube & Door

Laser cutting Traits

- Structural frames
- Flat pieces

Where? RPC





Project Roadmap WBS Draft

Mile stone 1:	Conceptual design	
task 1	Project Ideation	
task 2	Design reviewed	
Mile stone 2:	Movement system consolidated	
task 1	chassis and wheels built	
task 2	Optic sensor configured	
task 3	Path & position programmed	
Mile stone 3:	Compression system work as expected	
task 2	compression mechanism built	
task 2	Door movement mechanism built	
task 3	Compression task programmed	
Mile stone 4:	Scooping system work as expected	
task 1	Scooping mechanism CADed & build	
task 2	Object detection via distance sensor programmed	
task 3	Scoop repetition count programmed	
Mile stone 5:	All sensors configured and work together	
task 1	Testing prototype	
task 2	Demonstration prep	
task 3	Final report	



The background features a light gray color with decorative elements. In the top-left corner, there are circuit-like lines and a grid of dots. In the top-right corner, there is a large gear with a smaller gear inside it, and several lines extending downwards. In the bottom-left corner, there is another gear with a smaller gear inside it, and some lines extending from it.

THANK YOU
for
listening
me yap